“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 23

Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.3, 16.4 (fundamental theorem of line integrals and Green's theorem)

- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

ALVARO: Start the recording!

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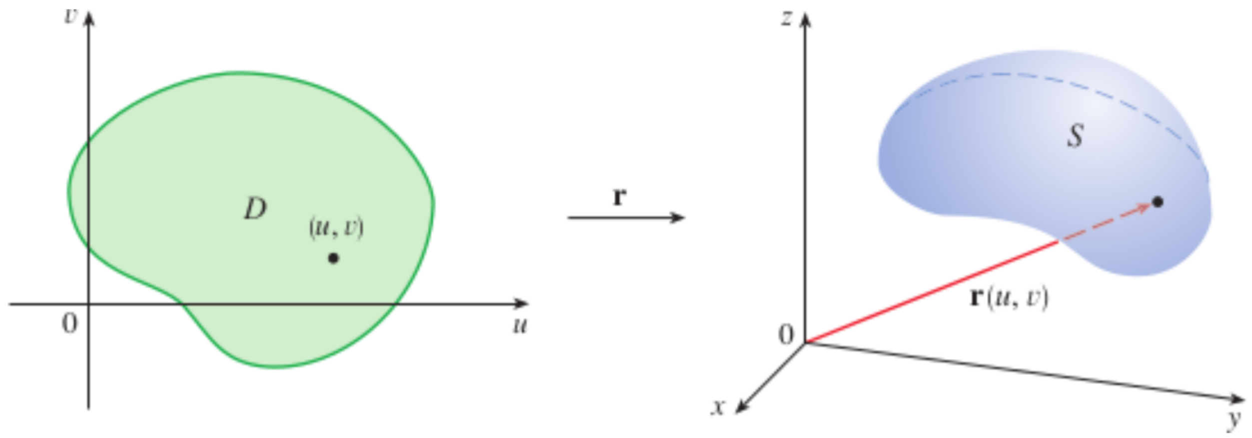
Parametric Surfaces

About Parametrized Surfaces

$$\mathbf{r}(u,v) = x(u,v)\,\mathbf{i} + y(u,v)\,\mathbf{j} + z(u,v)\,\mathbf{k}$$

A parametric surface



Example: Identify and sketch the surface with vector equation

$$r(u, v) = (1, u, u + v) = 1 \cdot i + u \cdot j + (u + v) \cdot k.$$

[Extra Space]

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$$r(u, v) = (u, u, u + v) = u \cdot i + u \cdot j + (u + v) \cdot k.$$

Example: Find a parametric equation for the segment of a cylinder

$$x^2 + y^2 = 4, \qquad 0 \leq z \leq 1.$$

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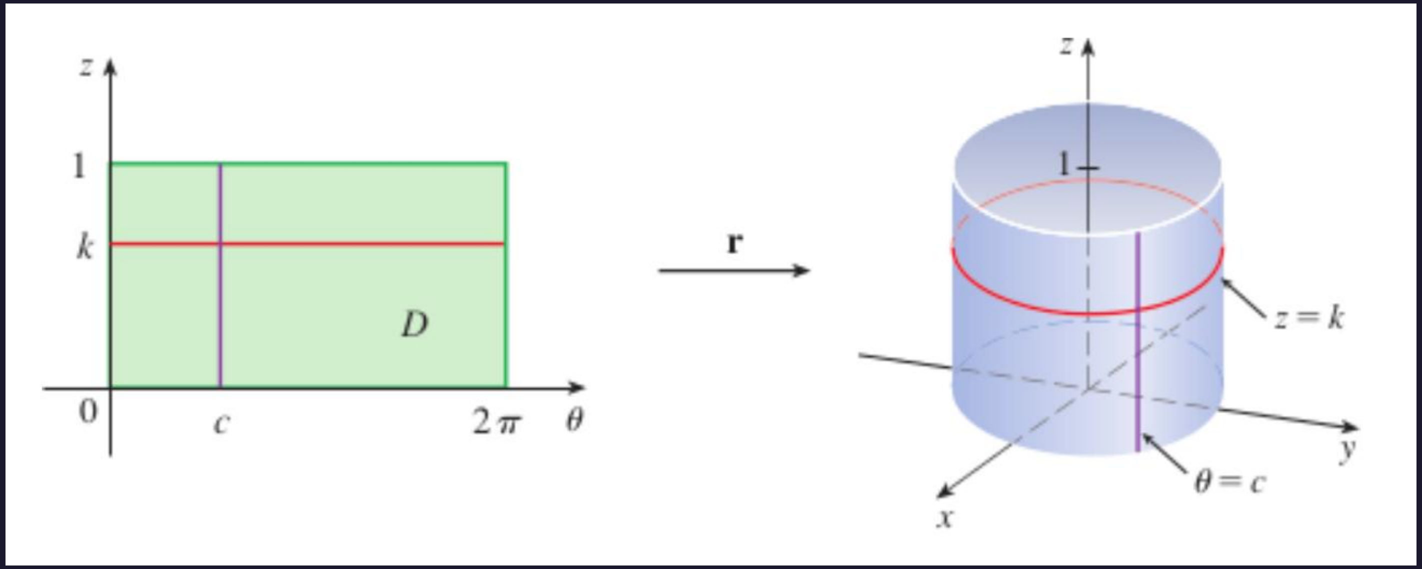

Example: Identify and sketch the surface with vector equation

$$r(u, v) = (2\cos(u), v, 2\sin(u))$$

[Extra Space]

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where $0 \leq u \leq \frac{\pi}{2}$, $0 \leq v \leq 3$.

[Extra Space]

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Example: Find a parametric equation for the elliptic paraboloid

$$z = x^2 + 2y^2.$$

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Parametric surfaces can be complicated!

$$\left(X(u,v), Y(u,v), Z(u,v)\right)$$

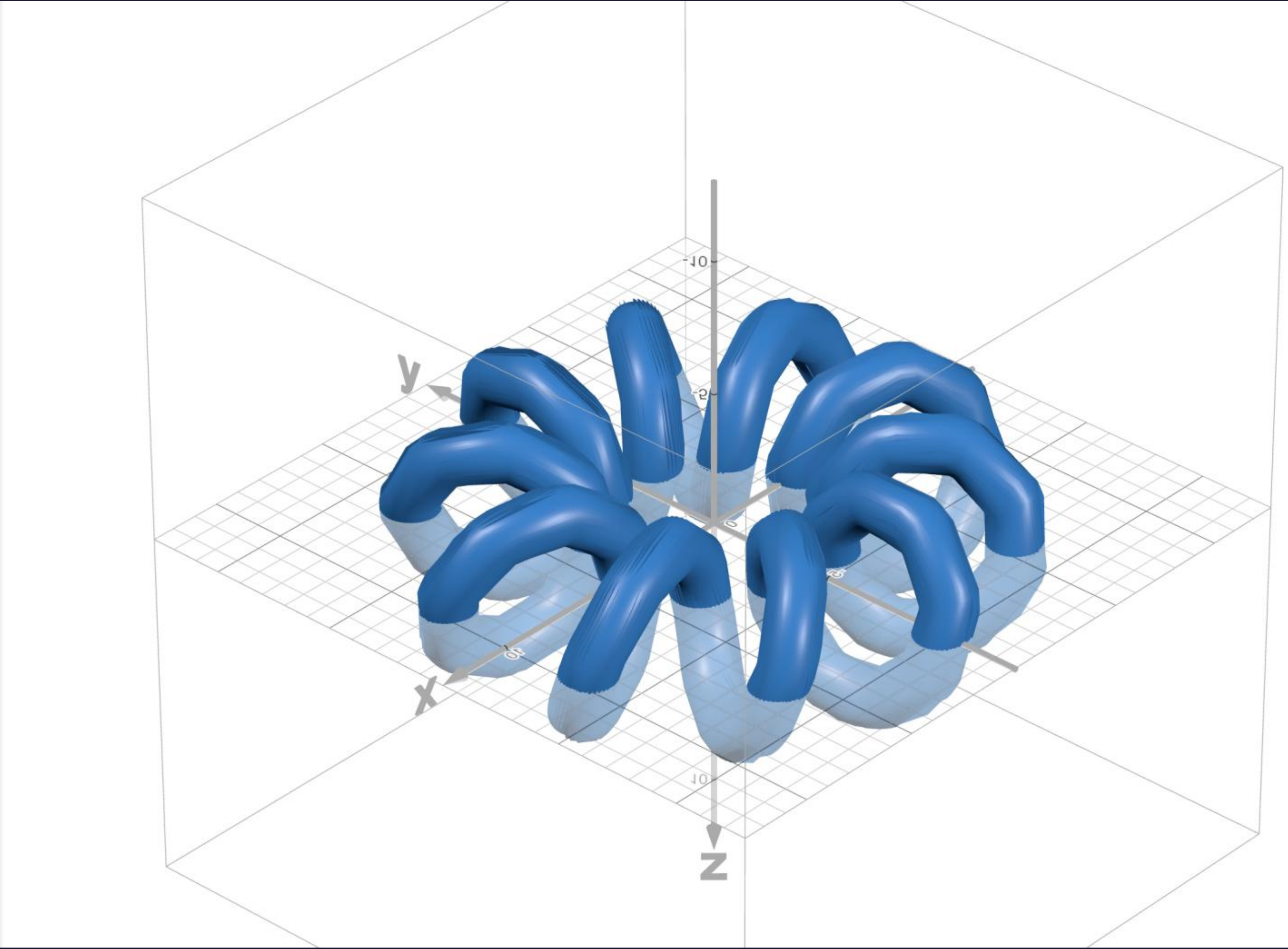
$0 \le u \le 1$

$0 \le v \le 1$

$$X(u,v) = \cos(\tau v)\cos(\tau u) + \left(\cos(10\tau v)\left(\sin(\tau u) + 3\right) + 7\right)\left(-\sin(\tau v)\right)$$

$$Y(u,v) = \sin(\tau v)\cos(\tau u) + \left(\cos(10\tau v)\left(\sin(\tau u) + 3\right) + 7\right)\left(\cos(\tau v)\right)$$

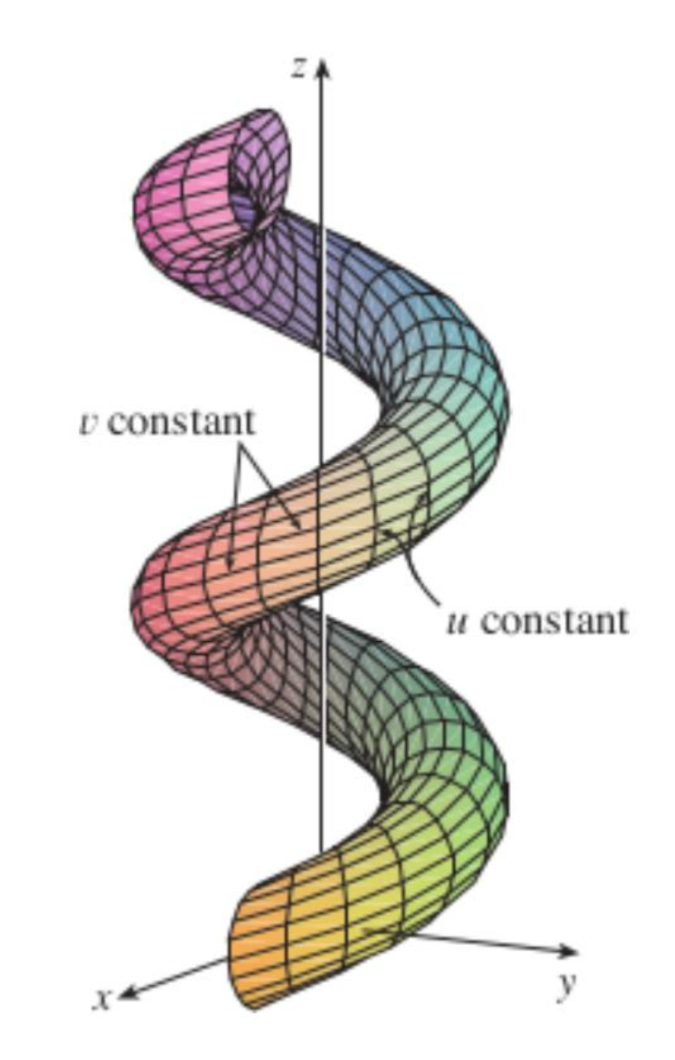
$$Z(u,v) = \left(\sin(10\tau v)\right)\left(\sin(\tau u) + 3\right)$$



Grid curves on parametric surfaces

Example: Sketch the "grid curves" with constant v

$$r(u,v) = \big((2+\sin(v))\cos(u), (2+\sin(v))\sin(u), u+\cos(v)\big)$$



[Extra Space]

Example: Sketch the "grid curves" with constant v

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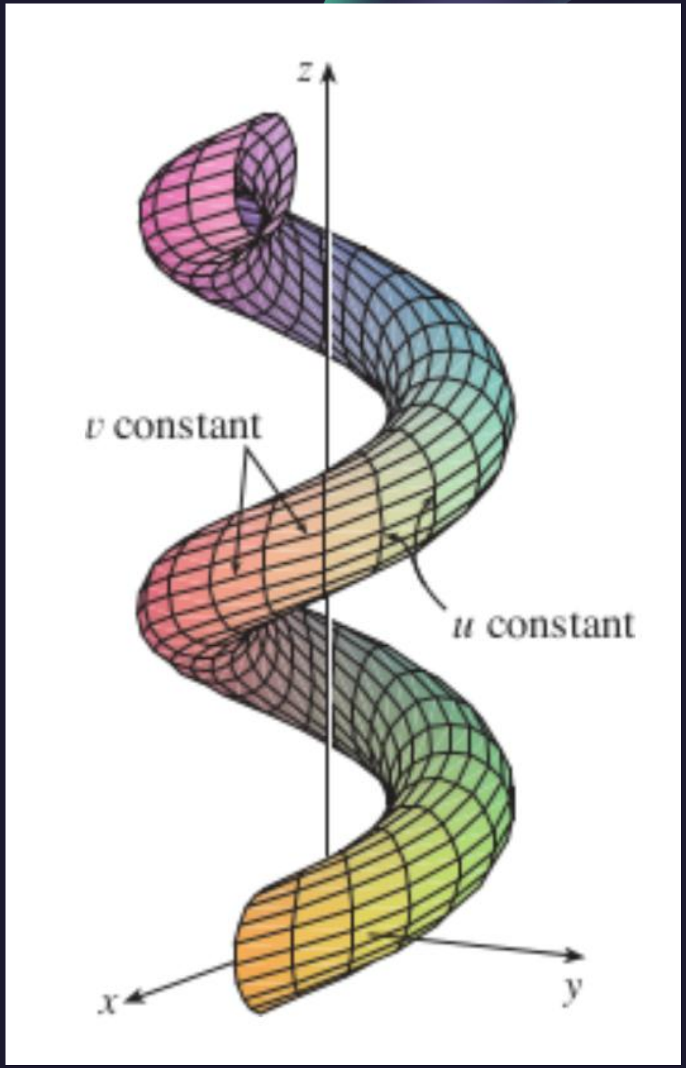
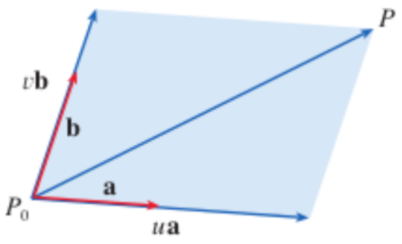


Example: Find a vector equation for the plane that passes through a point P and that contains two non-parallel vectors a and b.

[Extra Space]

Example: Find a vector equation for the plane that passes through a point P and that contains two non-parallel vectors a and b.

Figure 6



If we write $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$, $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then we can write the parametric equations of the plane through the point (x_0, y_0, z_0) as follows:

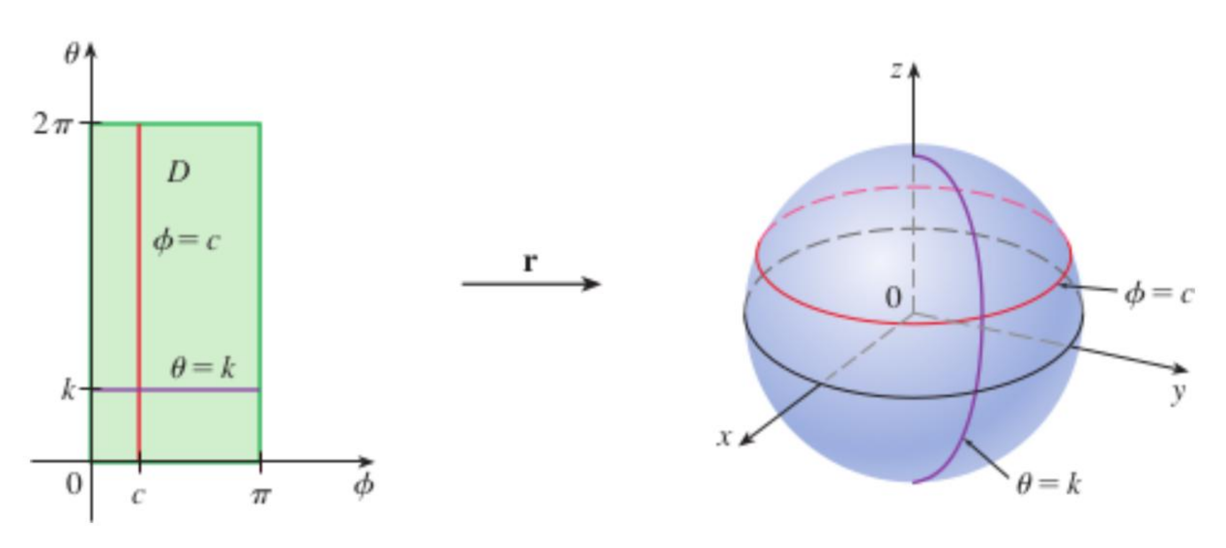
$$x = x_0 + ua_1 + vb_1 \qquad y = y_0 + ua_2 + vb_2 \qquad z = z_0 + ua_3 + vb_3$$

Example: Find a parametric equation for a sphere centered at (0,0,0) of radius R.

[Extra Space]

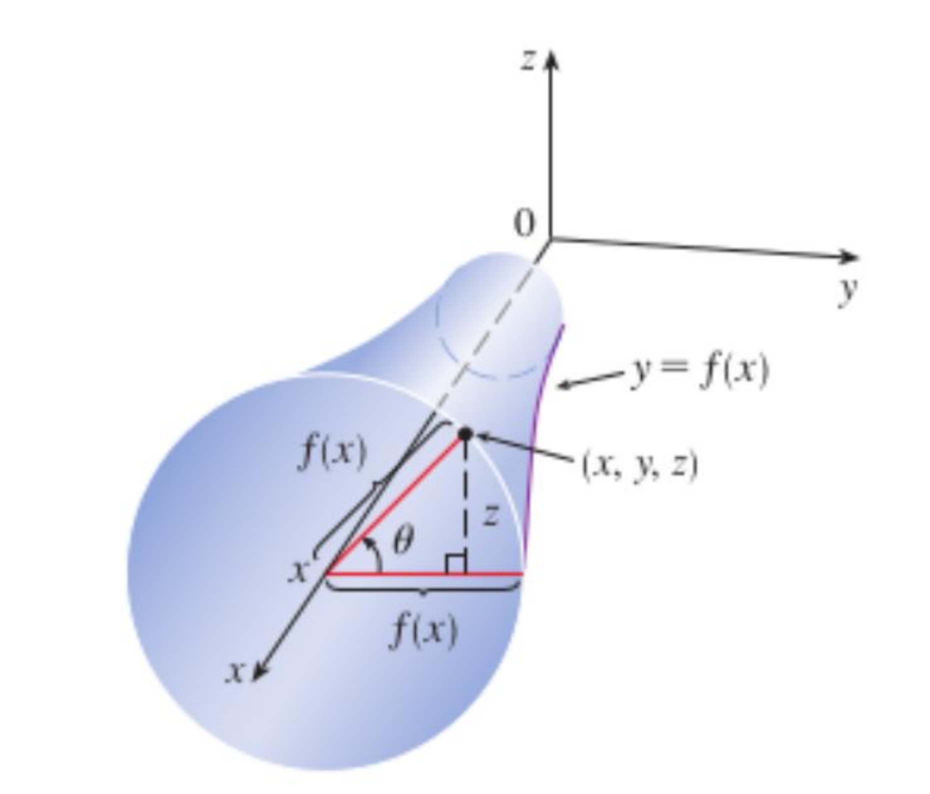
Example: Find a parametric equation for a sphere centered at (0,0,0) of radius R.

$$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$$



Surfaces of Revolution

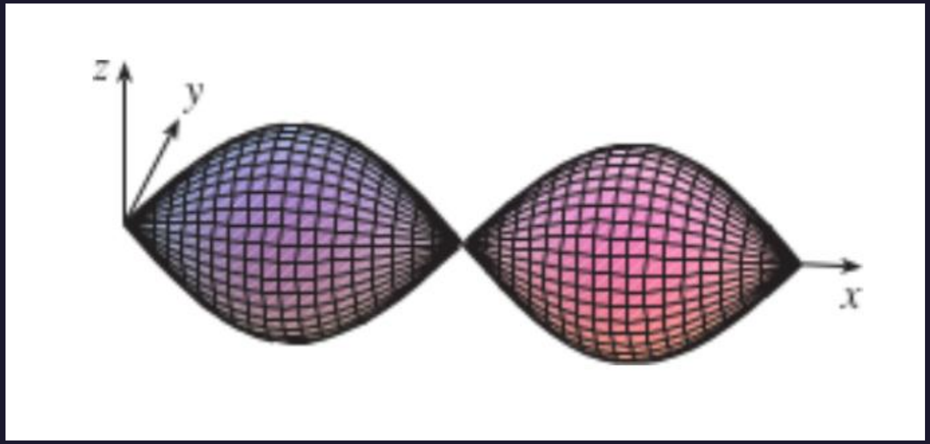
$$x = x \qquad y = f(x)\cos\theta \qquad z = f(x)\sin\theta$$



Example: Find a parametric equation for the surface that is obtained by rotating the curve $y = \sin(x)$, $0 \leq x \leq 2\pi$, about the x-axis.

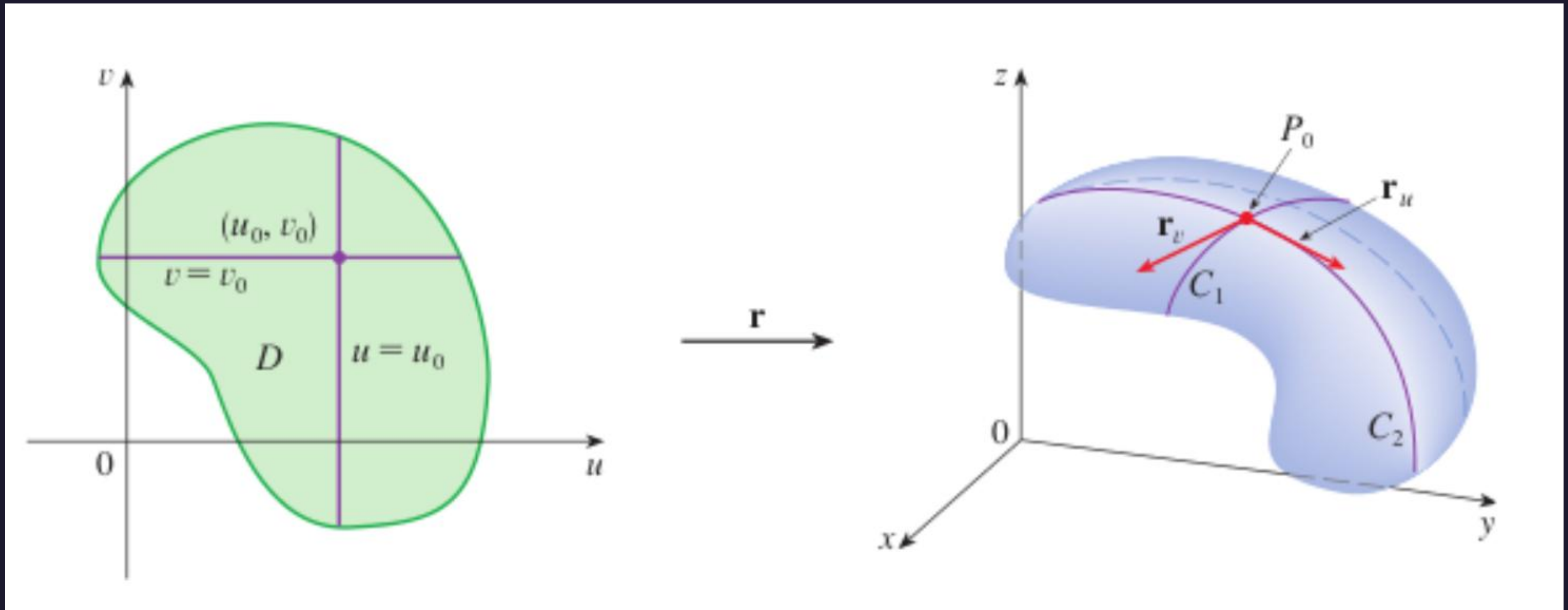
[Extra Space]

Example: Find a parametric equation for the surface that is obtained by rotating the curve $y = \sin(x)$, $0 \le x \le 2\pi$, about the x-axis.



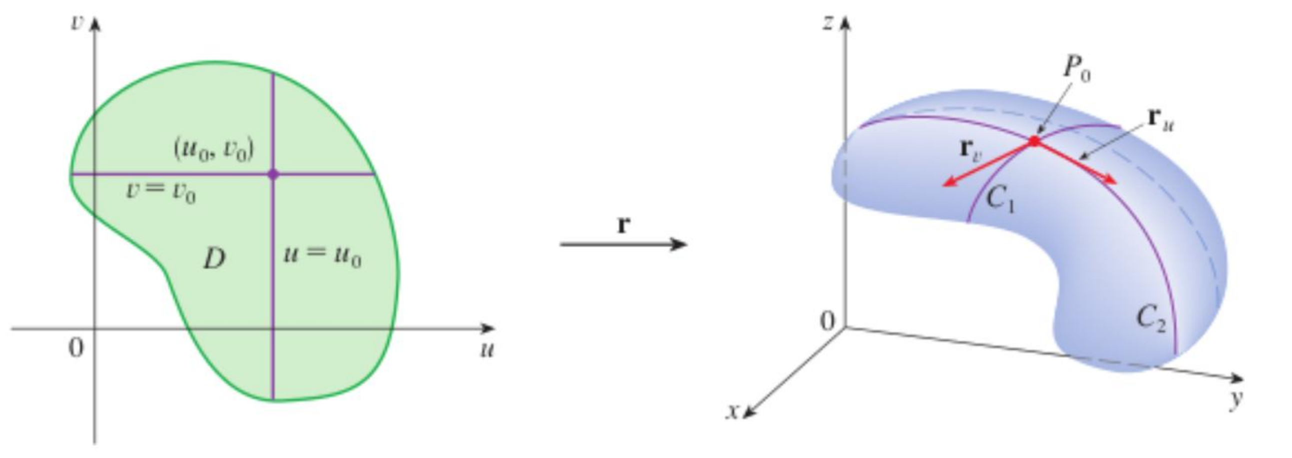
Tangent Planes

$$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k}$$



Tangent Planes

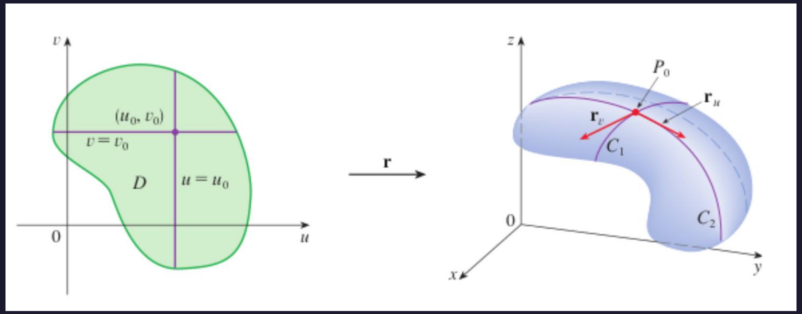
$$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k}$$



$$\mathbf{r}_u = \frac{\partial x}{\partial u}(u_0, v_0)\,\mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0)\,\mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0)\,\mathbf{k}$$

$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0, v_0)\,\mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0)\,\mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0)\,\mathbf{k}$$

Tangent Planes



$$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k}$$

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$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0, v_0)\,\mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0)\,\mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0)\,\mathbf{k}$$

If $\mathbf{r}_u \times \mathbf{r}_v$ is not $\mathbf{0}$, then the surface S is called **smooth** (it has no "corners"). For a smooth surface, the **tangent plane** is the plane that contains the tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$, and the vector $\mathbf{r}_u \times \mathbf{r}_v$ is a normal vector to the tangent plane.

Example: Find the equation of the tangent plane to the surface with parametric equations $x = u^2, y = v^2, z = u + 2v$ at the point P=(1,1,3).

[Extra Space]

Example: Find the equation of the tangent plane to the surface with parametric equations $x = u^2, y = v^2, z = u + 2v$ at the point P=(1,1,3).

Questions?

ALVARO: Start the recording!

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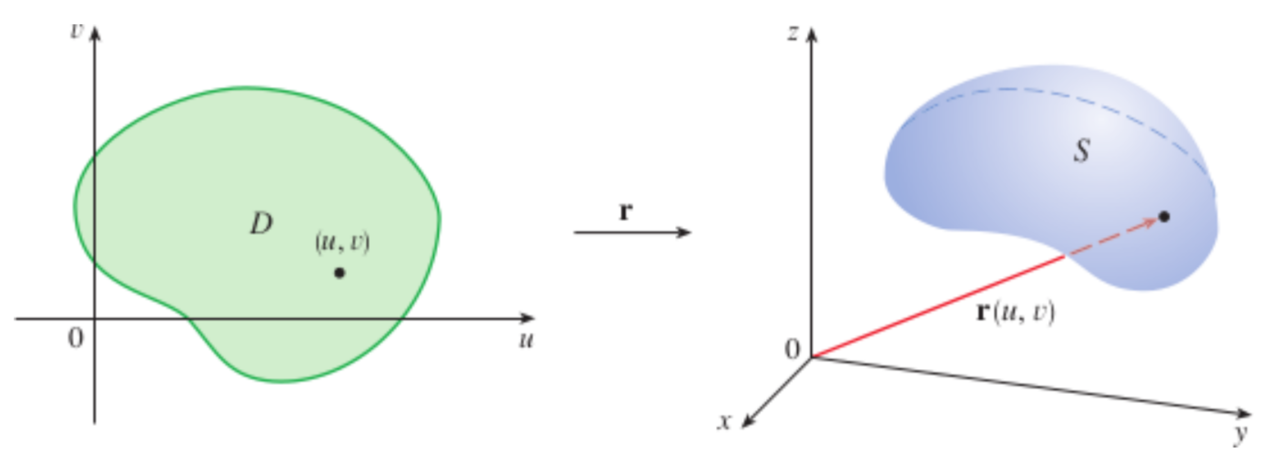
Surface Area

About Parametrized Surfaces and their Area

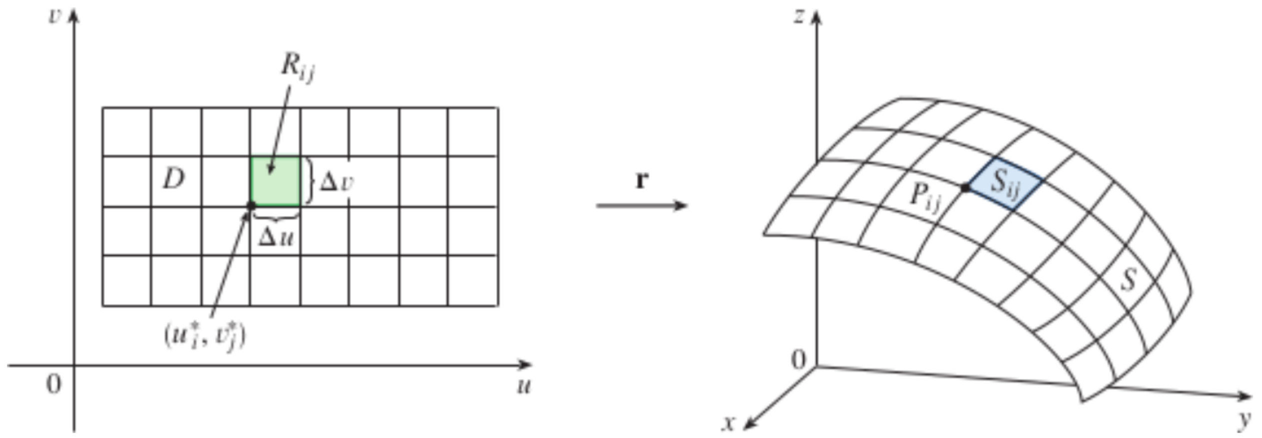
$$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k}$$

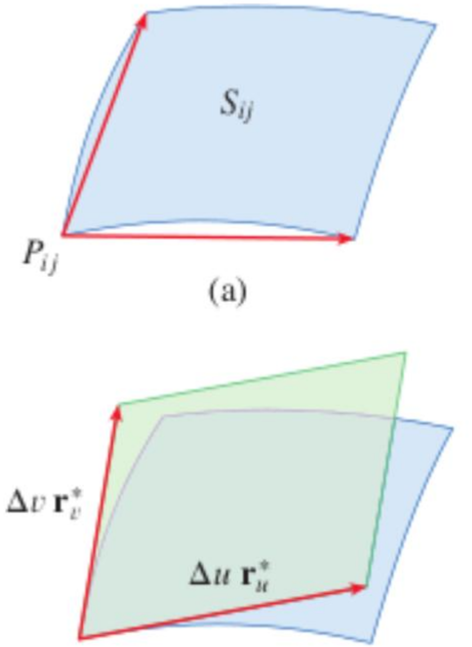
A parametric surface



About Parametrized Surfaces and their Area

The image of the subrectangle R_{ij} is the patch S_{ij}.





$$|(\Delta u\, \mathbf{r}_u^*) \times (\Delta v\, \mathbf{r}_v^*)| = |\mathbf{r}_u^* \times \mathbf{r}_v^*|\, \Delta u\, \Delta v$$

$$\sum_{i=1}^{m} \sum_{j=1}^{n} |\mathbf{r}_u^* \times \mathbf{r}_v^*|\, \Delta u\, \Delta v$$

About Parametrized Surfaces and their Area

If a smooth parametric surface S is given by the equation

$$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k} \qquad (u, v) \in D$$

and S is covered just once as (u, v) ranges throughout the parameter domain D, then the **surface area** of S is

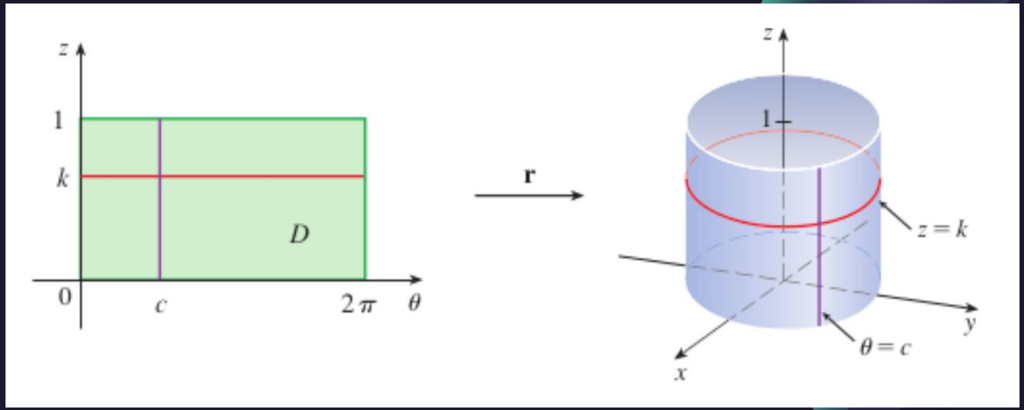
$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

where

$$\mathbf{r}_u = \frac{\partial x}{\partial u}\,\mathbf{i} + \frac{\partial y}{\partial u}\,\mathbf{j} + \frac{\partial z}{\partial u}\,\mathbf{k} \qquad \mathbf{r}_v = \frac{\partial x}{\partial v}\,\mathbf{i} + \frac{\partial y}{\partial v}\,\mathbf{j} + \frac{\partial z}{\partial v}\,\mathbf{k}$$

Example: Compute the surface area of the segment of a cylinder

$$x^2 + y^2 = 4, \qquad 0 \le z \le 1.$$



[Extra Space]

Example: Find a parametric equation for the segment of a cylinder

$$x^2 + y^2 = 4, \qquad 0 \leq z \leq 1.$$

Example: Compute the surface area of a sphere of radius R.

$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$

[Extra Space]

Example: Compute the surface area of a sphere of radius R.

$$\mathbf{r}_\phi \times \mathbf{r}_\theta = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial \phi} & \frac{\partial y}{\partial \phi} & \frac{\partial z}{\partial \phi} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a \cos \phi \cos \theta & a \cos \phi \sin \theta & -a \sin \phi \\ -a \sin \phi \sin \theta & a \sin \phi \cos \theta & 0 \end{vmatrix}$$

$$= a^2 \ \sin^2 \phi \cos \theta \, \mathbf{i} + a^2 \ \sin^2 \phi \sin \theta \, \mathbf{j} + a^2 \ \sin \phi \cos \phi \, \mathbf{k}$$

$$|\mathbf{r}_\phi \times \mathbf{r}_\theta| = \sqrt{a^4 \sin^4 \phi \cos^2 \theta + a^4 \sin^4 \phi \sin^2 \theta + a^4 \sin^2 \phi \cos^2 \phi}$$

$$= \sqrt{a^4 \sin^4 \phi + a^4 \sin^2 \phi \cos^2 \phi} = a^2 \sqrt{\sin^2 \phi} = a^2 \sin \phi$$

Surfaces Area of a Graph $z = f(x,y)$

Surfaces Area of a Graph (and Length of a Graph)

$$A(S) = \iint\limits_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9.

$$x = a \sin \phi \cos \theta \qquad y = a \sin \phi \sin \theta \qquad z = a \cos \phi$$

[Extra Space]

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9.

Surfaces Area of a Surface of Revolution

Surfaces Area of a Surface of Revolution

$$A = \iint\limits_D |\mathbf{r}_x \times \mathbf{r}_\theta| \; dA$$

$$= \int_0^{2\pi} \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \; dx \, d\theta$$

$$= 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \; dx$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9, as a surface of revolution.

$$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9, as a surface of revolution.

Questions?

Thank you

Until next time.